



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°3_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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NIVEAU: L3

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L'objectif de ce TD est de:

1. Surveiller les ressources système
2. Analyser les logs avec journalctl
3. Comprendre les logs applicatifs
4. Mettre en place une supervision simple
5. Détecter un incident
6. Mettre en place une supervision simple automatisée
7. Comprendre la rotation des logs

1. Supervision système en temps reel

```
Last login: Sun Mar 15 21:33:40 2026 from 10.79.242.160
lens@lens:~$ sudo apt install htop -y
htop
[sudo] password for lens:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
htop is already the newest version (3.3.0-4build1).
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.

[1]+  Stopped                  htop
lens@lens:~$ |
```

```
lens@lens: ~  
Command Prompt: lens@lens: ~  
ctrl+alt+1  
Mem[|||||] 1.6% Tasks: 25, 22 thr, 77 kthr; 1 runnin  
Swp[ 198M/1.92G] Load average: 0.03 0.02 0.00  
0K/2.00G] Uptime: 00:11:07  
  
Main I/O  
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command  
1019 lens 20 0 15128 7184 5184 S 0.8 0.4 0:00.84 sshd: len  
1 root 20 0 22132 13268 9512 S 0.0 0.7 0:03.46 /sbin/ini  
295 root 19 -1 50468 17272 16124 S 0.0 0.9 0:00.32 /usr/lib/  
350 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.07 /sbin/mul  
363 root 20 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/mul  
364 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/mul  
365 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/mul  
366 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/mul  
367 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.07 /sbin/mul  
368 root RT 0 282M 27452 8760 S 0.0 1.4 0:00.00 /sbin/mul  
372 root 20 0 29076 7820 5024 S 0.0 0.4 0:00.23 /usr/lib/  
519 systemd-ne 20 0 19012 9480 8316 S 0.0 0.5 0:00.15 /usr/lib/  
533 systemd-re 20 0 21592 13036 10744 S 0.0 0.6 0:00.17 /usr/lib/  
536 systemd-ti 20 0 91028 7932 6960 S 0.0 0.4 0:00.08 /usr/lib/  
568 systemd-ti 20 0 91028 7932 6960 S 0.0 0.4 0:00.00 /usr/lib/  
651 messagebus 20 0 9808 5568 4688 S 0.0 0.3 0:00.11 @dbus-dae  
655 polkitd 20 0 374M 9956 7528 S 0.0 0.5 0:00.16 /usr/lib/  
662 root 20 0 18208 8872 7776 S 0.0 0.4 0:00.11 /usr/lib/  
666 root 20 0 457M 13612 11408 S 0.0 0.7 0:00.20 /usr/libe  
671 syslog 20 0 217M 6156 4608 S 0.0 0.3 0:00.06 /usr/sbin  
684 root 20 0 107M 23132 13680 S 0.0 1.1 0:00.40 /usr/bin/  
688 root 20 0 457M 13612 11408 S 0.0 0.7 0:00.01 /usr/libe  
690 root 20 0 457M 13612 11408 S 0.0 0.7 0:00.00 /usr/libe  
692 polkitd 20 0 374M 9956 7528 S 0.0 0.5 0:00.02 /usr/lib/  
693 polkitd 20 0 374M 9956 7528 S 0.0 0.5 0:00.00 /usr/lib/  
694 root 20 0 457M 13612 11408 S 0.0 0.7 0:00.00 /usr/libe  
695 polkitd 20 0 374M 9956 7528 S 0.0 0.5 0:00.00 /usr/lib/  
706 syslog 20 0 217M 6156 4608 S 0.0 0.3 0:00.00 /usr/sbin  
707 syslog 20 0 217M 6156 4608 S 0.0 0.3 0:00.00 /usr/sbin  
708 syslog 20 0 217M 6156 4608 S 0.0 0.3 0:00.01 /usr/sbin  
709 root 20 0 382M 12920 10848 S 0.0 0.6 0:00.26 /usr/sbin  
712 root 20 0 457M 13612 11408 S 0.0 0.7 0:00.00 /usr/libe  
714 root 20 0 6824 2912 2664 S 0.0 0.1 0:00.02 /usr/sbin  
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice - F8Nice + F9Kill F10Q
```

```
lens@lens:~$ df -h  
Filesystem Size Used Avail Use% Mounted on  
tmpfs 197M 1.1M 196M 1% /run  
/dev/mapper/ubuntu--vg-ubuntu--lv 12G 4.8G 6.0G 45% /  
tmpfs 985M 0 985M 0% /dev/shm  
tmpfs 5.0M 0 5.0M 0% /run/lock  
/dev/sda2 2.0G 104M 1.7G 6% /boot  
tmpfs 197M 12K 197M 1% /run/user/1000
```

2. Analyse des logs système

```
lens@lens:~$ journalctl
Mar 08 17:17:25 lens kernel: Linux version 6.8.0-101-generic (buildd@lcy02->
Mar 08 17:17:25 lens kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-101-ge>
Mar 08 17:17:25 lens kernel: KERNEL supported cpus:
Mar 08 17:17:25 lens kernel: Intel GenuineIntel
Mar 08 17:17:25 lens kernel: AMD AuthenticAMD
Mar 08 17:17:25 lens kernel: Hygon HygonGenuine
Mar 08 17:17:25 lens kernel: Centaur CentaurHauls
Mar 08 17:17:25 lens kernel: zhaoxin Shanghai
Mar 08 17:17:25 lens kernel: [Firmware Bug]: TSC doesn't count with P0 freq>
Mar 08 17:17:25 lens kernel: BIOS-provided physical RAM map:
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x0000000000009fc00-0x0000000000>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x000000000000f0000-0x0000000000>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x00000000000100000-0x0000000007>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x0000000007ffff000-0x0000000007>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x000000000fec00000-0x000000000f>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x000000000fee00000-0x000000000f>
Mar 08 17:17:25 lens kernel: BIOS-e820: [mem 0x000000000fffc0000-0x000000000f>
Mar 08 17:17:25 lens kernel: NX (Execute Disable) protection: active
Mar 08 17:17:25 lens kernel: APIC: Static calls initialized
Mar 08 17:17:25 lens kernel: SMBIOS 2.5 present.
Mar 08 17:17:25 lens kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS >
Mar 08 17:17:25 lens kernel: Hypervisor detected: KVM
Mar 08 17:17:25 lens kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Mar 08 17:17:25 lens kernel: kvm-clock: using sched offset of 9403700493055>
Mar 08 17:17:25 lens kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff>
Mar 08 17:17:25 lens kernel: tsc: Detected 1996.199 MHz processor
Mar 08 17:17:25 lens kernel: e820: update [mem 0x000000000-0x000000fff] usabl>
Mar 08 17:17:25 lens kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Mar 08 17:17:25 lens kernel: last_pfn = 0x7fff0 max_arch_pfn = 0x400000000
Mar 08 17:17:25 lens kernel: MTRR map: 3 entries (3 fixed + 0 variable; max>
Mar 08 17:17:25 lens kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC >
Mar 08 17:17:25 lens kernel: CPU MTRRs all blank - virtualized system.
Mar 08 17:17:25 lens kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fb>
Mar 08 17:17:25 lens kernel: RAMDISK: [mem 0x2ee57000-0x33722fff]
Mar 08 17:17:25 lens kernel: ACPI: Early table checksum verification disabl>
Mar 08 17:17:25 lens kernel: ACPI: RSDP 0x000000000000E0000 000024 (v02 VBOX>
Mar 08 17:17:25 lens kernel: ACPI: XSDT 0x0000000007FFF0030 00003C (v01 VBOX>
```



```
lens@lens: ~  
There are stopped jobs.  
lens@lens:~$ journalctl -xe  
Defined-By: systemd  
Support: http://www.ubuntu.com/support  
Documentation: sd-login(3)  
  
A new session with the ID 4 has been created for the user lens.  
  
The leading process of the session is 964.  
Mar 21 18:25:51 lens systemd[1]: Started session-4.scope - Session 4 of Use  
Subject: A start job for unit session-4.scope has finished successfully  
Defined-By: systemd  
Support: http://www.ubuntu.com/support  
  
A start job for unit session-4.scope has finished successfully.  
  
The job identifier is 995.  
Mar 21 18:27:35 lens sudo[1029]: lens : TTY=pts/0 ; PWD=/home/lens ; US  
Mar 21 18:27:35 lens sudo[1029]: pam_unix(sudo:session): session opened for  
Mar 21 18:27:38 lens sudo[1029]: pam_unix(sudo:session): session closed for  
Mar 21 18:30:01 lens systemd[1]: Starting sysstat-collect.service - system  
Subject: A start job for unit sysstat-collect.service has begun execution  
Defined-By: systemd  
Support: http://www.ubuntu.com/support  
  
A start job for unit sysstat-collect.service has begun execution.  
  
The job identifier is 1107.  
Mar 21 18:30:01 lens systemd[1]: sysstat-collect.service: Deactivated succe  
Subject: Unit succeeded  
Defined-By: systemd  
Support: http://www.ubuntu.com/support  
  
The unit sysstat-collect.service has successfully entered the 'dead' sta  
Mar 21 18:30:01 lens systemd[1]: Finished sysstat-collect.service - system  
Subject: A start job for unit sysstat-collect.service has finished succe  
Defined-By: systemd  
Support: http://www.ubuntu.com/support  
  
A start job for unit sysstat-collect.service has finished successfully.
```

```

lens@lens: ~$ journalctl -b
Mar 21 18:16:58 lens kernel: Linux version 6.8.0-101-generic (builddd@lcy02->
Mar 21 18:16:58 lens kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-101-ge>
Mar 21 18:16:58 lens kernel: KERNEL supported cpus:
Mar 21 18:16:58 lens kernel:   Intel GenuineIntel
Mar 21 18:16:58 lens kernel:   AMD AuthenticAMD
Mar 21 18:16:58 lens kernel:   Hygon HygonGenuine
Mar 21 18:16:58 lens kernel:   Centaur CentaurHauls
Mar 21 18:16:58 lens kernel:   zhaoxin   Shanghai
Mar 21 18:16:58 lens kernel: [Firmware Bug]: TSC doesn't count with P0 freq>
Mar 21 18:16:58 lens kernel: BIOS-provided physical RAM map:
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x00000000000009fc00-0x000000000>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x0000000000000f0000-0x000000000>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x000000000000100000-0x000000000>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x0000000000007fff0000-0x000000000>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x000000000fec000000-0x000000000f>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x000000000fee000000-0x000000000f>
Mar 21 18:16:58 lens kernel: BIOS-e820: [mem 0x000000000fffc00000-0x000000000f>
Mar 21 18:16:58 lens kernel: NX (Execute Disable) protection: active
Mar 21 18:16:58 lens kernel: APIC: Static calls initialized
Mar 21 18:16:58 lens kernel: SMBIOS 2.5 present.
Mar 21 18:16:58 lens kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS >
Mar 21 18:16:58 lens kernel: Hypervisor detected: KVM
Mar 21 18:16:58 lens kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Mar 21 18:16:58 lens kernel: kvm-clock: using sched offset of 9940823707 cy>
Mar 21 18:16:58 lens kernel: clocksource: kvm-clock: mask: 0xffffffffffffff>
Mar 21 18:16:58 lens kernel: tsc: Detected 1996.198 MHz processor
Mar 21 18:16:58 lens kernel: e820: update [mem 0x000000000-0x00000ffff] usabl>
Mar 21 18:16:58 lens kernel: e820: remove [mem 0x000a00000-0x000ffffff] usable>
Mar 21 18:16:58 lens kernel: last_pfn = 0x7fff0 max_arch_pfn = 0x400000000
Mar 21 18:16:58 lens kernel: MTRR map: 3 entries (3 fixed + 0 variable; max>
Mar 21 18:16:58 lens kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC >
Mar 21 18:16:58 lens kernel: CPU MTRRs all blank - virtualized system.
Mar 21 18:16:58 lens kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fb>
Mar 21 18:16:58 lens kernel: RAMDISK: [mem 0x2ee57000-0x33722fff]
Mar 21 18:16:58 lens kernel: ACPI: Early table checksum verification disabl>
Mar 21 18:16:58 lens kernel: ACPI: RSDP 0x000000000000E0000 000024 (v02 VBOX>
Mar 21 18:16:58 lens kernel: ACPI: XSDT 0x0000000007FFF0030 00003C (v01 VBOX>
Mar 21 18:16:58 lens kernel: ACPI: FACP 0x0000000007FFF00F0 0000F4 (v04 VBOX>
Mar 21 18:16:58 lens kernel: ACPI: DSDT 0x0000000007FFF02E0 002353 (v02 VBOX>

```



```

lens@lens: ~
lens@lens:~$ journalctl -p err
Mar 08 17:17:30 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seem>
Mar 08 17:17:30 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This config>
Mar 08 17:17:30 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swit>
Mar 08 17:22:54 lens login[905]: PAM unable to dlopen(pam_lastlog.so): /usr>
Mar 08 17:22:54 lens login[905]: PAM adding faulty module: pam_lastlog.so
-- Boot 911a092abdae42d1a90d7576f9a0b9d4 --
Mar 08 17:59:44 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seem>
Mar 08 17:59:44 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This config>
Mar 08 17:59:44 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swit>
Mar 08 18:01:21 lens login[747]: PAM unable to dlopen(pam_lastlog.so): /usr>
Mar 08 18:01:21 lens login[747]: PAM adding faulty module: pam_lastlog.so
Mar 08 19:16:15 lens systemd[1]: systemd-journald.service: Watchdog timeout>
-- Boot f77dc200a7a9455d8eee9846b4befa7f --
Mar 15 14:14:37 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seem>
Mar 15 14:14:37 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This config>
Mar 15 14:14:37 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swit>
Mar 15 15:26:38 lens kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 8>
Mar 15 15:30:39 lens kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 4>
Mar 15 15:55:09 lens systemd[1]: systemd-journald.service: Watchdog timeout>
Mar 15 15:56:39 lens login[803]: PAM unable to dlopen(pam_lastlog.so): /usr>
Mar 15 15:56:39 lens login[803]: PAM adding faulty module: pam_lastlog.so
-- Boot 82d336929d8344d4b0c577691f5ffa12 --
Mar 15 16:06:59 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seem>
Mar 15 16:06:59 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This config>
Mar 15 16:06:59 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swit>
Mar 15 16:09:00 lens login[739]: PAM unable to dlopen(pam_lastlog.so): /usr>
Mar 15 16:09:00 lens login[739]: PAM adding faulty module: pam_lastlog.so
Mar 15 16:24:57 lens kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 5>
Mar 15 17:15:45 lens systemd[924]: Failed to create timezone change event s>
Mar 15 17:19:56 lens kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 2>
-- Boot 6a8ffc35fd364b699afa16882b78e0a8 --
Mar 15 20:06:41 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seem>
Mar 15 20:06:41 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This config>
Mar 15 20:06:41 lens kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please swit>
Mar 15 20:06:59 lens login[752]: PAM unable to dlopen(pam_lastlog.so): /usr>
Mar 15 20:06:59 lens login[752]: PAM adding faulty module: pam_lastlog.so
Mar 15 21:18:27 lens adduser[2940]: Only root may add a user or group to th>
Mar 15 21:18:39 lens adduser[2943]: The user 'backupuser' already exists.
Mar 15 21:19:23 lens adduser[2947]: The user 'backupuser' already exists.
Mar 15 21:48:41 lens sudo[3631]: backupuser : user NOT in sudoers ; TTY=pts>

```

```

lens@lens:~$ journalctl -u nginx
Mar 21 18:51:47 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 21 18:51:47 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
-- Boot 298bfeebc0064b1182bc9eb25483028a --
Mar 21 19:06:44 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 21 19:06:45 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
lens@lens:~$

```

3. Logs du serveur web

```

lens@lens:~$ sudo systemctl restart nginx
lens@lens:~$ tail -f /var/log/nginx/access.log
::1 - - [21/Mar/2026:19:25:52 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.5.0"
::1 - - [21/Mar/2026:19:26:03 +0000] "GET /404.html HTTP/1.1" 404 162 "-" "curl/8.5.0"
^Z
[6]+ Stopped                  tail -f /var/log/nginx/access.log
lens@lens:~$ tail -f /var/log/nginx/error.log
2026/03/21 18:51:47 [notice] 1355#1355: using inherited sockets from "5;6;"
^Z
[7]+ Stopped                  tail -f /var/log/nginx/error.log
lens@lens:~$ grep 404 /var/log/nginx/access.log
::1 - - [21/Mar/2026:19:26:03 +0000] "GET /404.html HTTP/1.1" 404 162 "-" "curl/8.5.0"
lens@lens:~$ |

```

4. Simulation d'incident

```

lens@lens: ~
url/8.5.0"
lens@lens:~$ sudo systemctl stop nginx
lens@lens:~$ systemctl status nginx
○ nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset>
   Active: inactive (dead) since Sat 2026-03-21 19:30:22 UTC; 19s ago
   Duration: 4min 4.777s
   Docs: man:nginx(8)
   Process: 1130 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_p>
   Process: 1135 ExecStart=/usr/sbin/nginx -g daemon on; master_process on>
   Process: 1149 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry Q>
   Main PID: 1136 (code=exited, status=0/SUCCESS)
   CPU: 62ms

Mar 21 19:26:17 lens systemd[1]: Starting nginx.service - A high performanc>
Mar 21 19:26:17 lens systemd[1]: Started nginx.service - A high performanc>
Mar 21 19:30:22 lens systemd[1]: Stopping nginx.service - A high performanc>
Mar 21 19:30:22 lens systemd[1]: nginx.service: Deactivated successfully.
Mar 21 19:30:22 lens systemd[1]: Stopped nginx.service - A high performanc>
lines 1-16/16 (END)
[8]+ Stopped                  systemctl status nginx

lens@lens:~$ journalctl -u nginx -n 20
Mar 21 19:06:44 lens systemd[1]: Starting nginx.service - A high performanc>
Mar 21 19:06:45 lens systemd[1]: Started nginx.service - A high performanc>
Mar 21 19:20:59 lens systemd[1]: Stopping nginx.service - A high performanc>
Mar 21 19:20:59 lens systemd[1]: nginx.service: Deactivated successfully.
Mar 21 19:20:59 lens systemd[1]: Stopped nginx.service - A high performanc>
Mar 21 19:20:59 lens systemd[1]: Starting nginx.service - A high performanc>
Mar 21 19:20:59 lens systemd[1]: Started nginx.service - A high performanc>
Mar 21 19:24:19 lens systemd[1]: Stopping nginx.service - A high performanc>
Mar 21 19:24:19 lens systemd[1]: nginx.service: Deactivated successfully.
Mar 21 19:24:19 lens systemd[1]: Stopped nginx.service - A high performanc>
Mar 21 19:24:26 lens systemd[1]: Starting nginx.service - A high performanc>
Mar 21 19:24:26 lens systemd[1]: Started nginx.service - A high performanc>
Mar 21 19:26:17 lens systemd[1]: Stopping nginx.service - A high performanc>
Mar 21 19:26:17 lens systemd[1]: nginx.service: Deactivated successfully.
Mar 21 19:26:17 lens systemd[1]: Stopped nginx.service - A high performanc>
Mar 21 19:26:17 lens systemd[1]: Starting nginx.service - A high performanc>
Mar 21 19:26:17 lens systemd[1]: Started nginx.service - A high performanc>

```



```
lens@lens: ~  
lens@lens:~$ journalctl -u nginx -n 20  
Mar 21 19:06:44 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:06:45 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:20:59 lens systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:20:59 lens systemd[1]: nginx.service: Deactivated successfully.  
Mar 21 19:20:59 lens systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:20:59 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:20:59 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:24:19 lens systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:24:19 lens systemd[1]: nginx.service: Deactivated successfully.  
Mar 21 19:24:19 lens systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:24:26 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:24:26 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:26:17 lens systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:26:17 lens systemd[1]: nginx.service: Deactivated successfully.  
Mar 21 19:26:17 lens systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:26:17 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:26:17 lens systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:30:22 lens systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server  
Mar 21 19:30:22 lens systemd[1]: nginx.service: Deactivated successfully.  
Mar 21 19:30:22 lens systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server  
[9]+ Stopped journalctl -u nginx -n 20  
lens@lens:~$ sudo systemctl start nginx  
lens@lens:~$ systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)  
   Active: active (running) since Sat 2026-03-21 19:31:48 UTC; 18s ago  
     Docs: man:nginx(8)  
  Process: 1162 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on;  
  Process: 1164 ExecStart=/usr/sbin/nginx -g daemon on; master_process on;  
 Main PID: 1165 (nginx)  
    Tasks: 2 (limit: 2263)  
  Memory: 1.7M (peak: 1.9M)  
     CPU: 47ms  
   CGroup: /system.slice/nginx.service  
           └─1165 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;  
             └─1166 "nginx: worker process"  
Mar 21 19:31:47 lens systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server
```

5. Analyse d'activité suspecte

```
</html>  
lens@lens:~$ grep "/admin" /var/log/nginx/access.log  
::1 - - [21/Mar/2026:19:44:36 +0000] "GET /admin HTTP/1.1" 404 162 "-" "curl  
/8.5.0"  
lens@lens:~$ |
```

6. Installer et configurer Logrotate

```
lens@lens:~$ sudo apt install logrotate -y
[sudo] password for lens:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
logrotate is already the newest version (3.21.0-2build1).
logrotate set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.
lens@lens:~$ cat /etc/logrotate.conf
# see "man logrotate" for details

# global options do not affect preceding include directives

# rotate log files weekly
weekly

# use the adm group by default, since this is the owning group
# of /var/log/.
su root adm

# keep 4 weeks worth of backlogs
rotate 4

# create new (empty) log files after rotating old ones
create

# use date as a suffix of the rotated file
#dateext

# uncomment this if you want your log files compressed
#compress

# packages drop log rotation information into this directory
include /etc/logrotate.d

# system-specific logs may also be configured here.
```

```
lens@lens: ~  
GNU nano 7.2 /etc/logrotate.d/devcloud *  
/var/log/monitor.log {  
daily  
rotate 7  
compress  
missingok  
notifempty  
create 640 root adm  
}  
|  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^J Justify
```



```

lens@lens:~$ ls /etc/logrotate.d/
alternatives  bootlog    devcloud  rsyslog    unattended-upgrades
appport       btmp       dpkg      ubuntu-pro-client  wtmp
apt           cloud-init nginx      ufw

lens@lens:~$ sudo nano /etc/logrotate.d/devcloud
lens@lens:~$ sudo logrotate -d /etc/logrotate.d/devcloud
warning: logrotate in debug mode does nothing except printing debug messages
! Consider using verbose mode (-v) instead if this is not what you want.

reading config file /etc/logrotate.d/devcloud
Reading state from file: /var/lib/logrotate/status
Allocating hash table for state file, size 64 entries
Creating new state

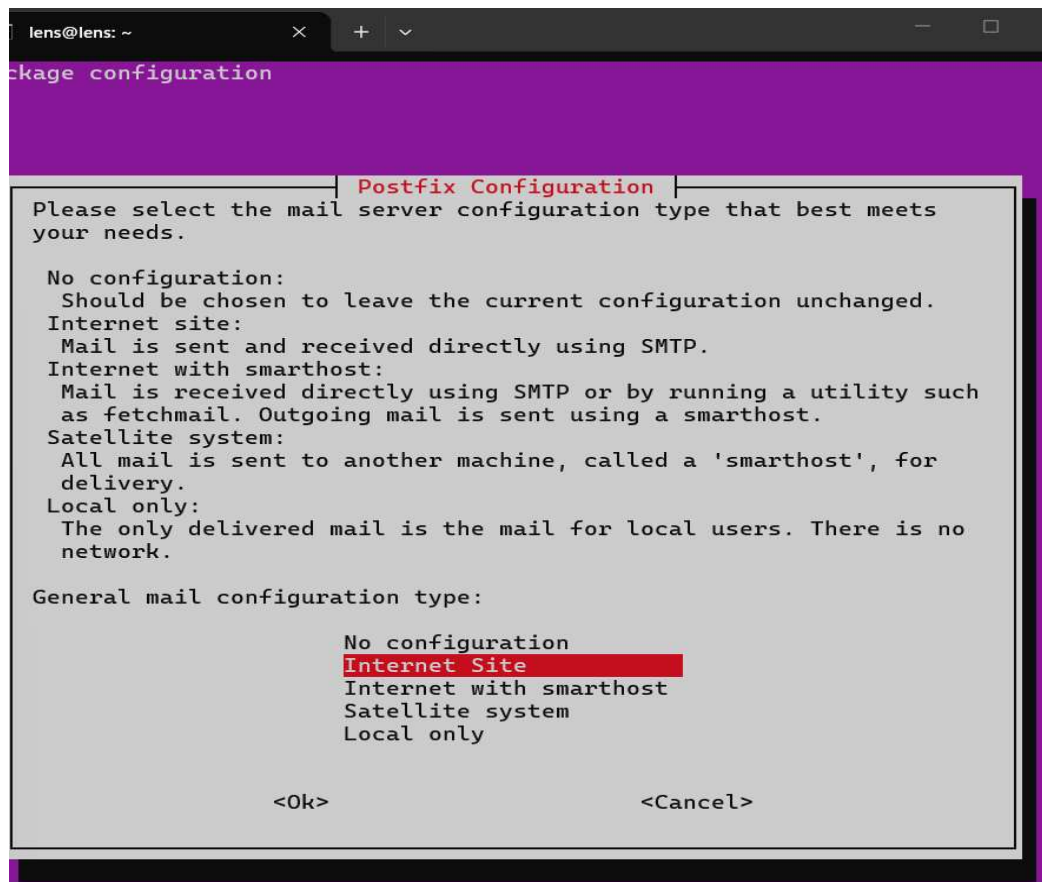
Handling 1 logs

rotating pattern: /var/log/monitor.log after 1 days (7 rotations)
empty log files are not rotated, old logs are removed
considering log /var/log/monitor.log
log /var/log/monitor.log does not exist -- skipping
lens@lens:~$ sudo logrotate -f /etc/logrotate.d/devcloud
lens@lens:~$

```

7. Installer un outil de supervision

Avec Nagios

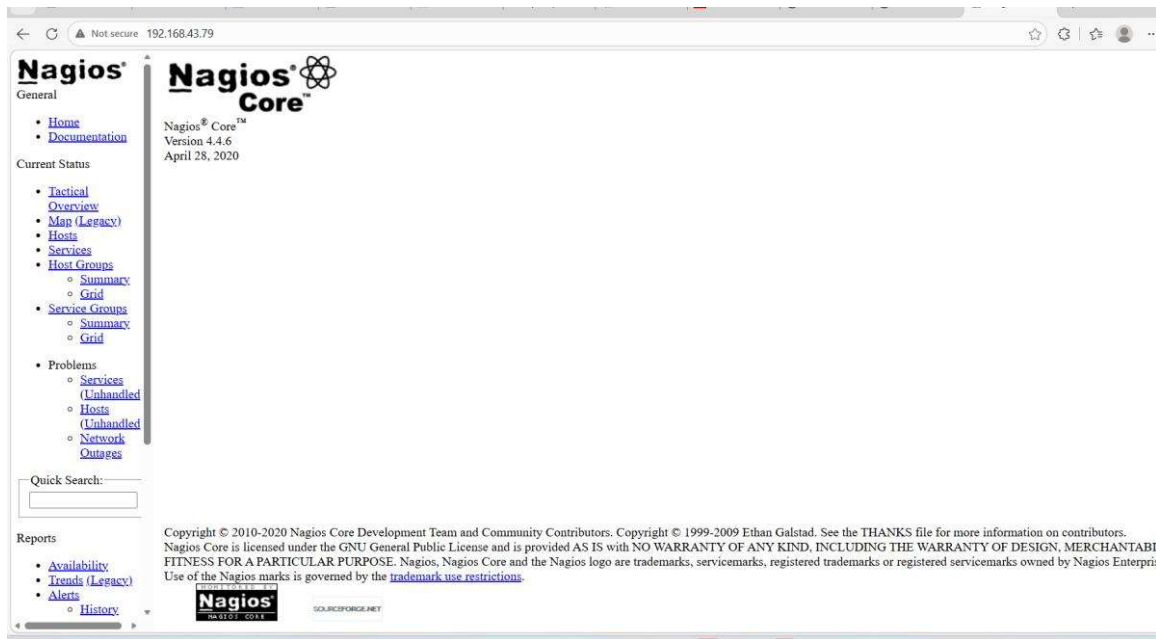


```

lens@lens: ~
[5]+ Stopped                                sudo systemctl status nagios4
lens@lens:~$ sudo systemctl restart nagios4
lens@lens:~$ sudo apt install nagios4 nagios-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'monitoring-plugins' instead of 'nagios-plugins'
nagios4 is already the newest version (4.4.6-4ubuntu0.24.04.1).
monitoring-plugins is already the newest version (2.3.5-1ubuntu3).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
lens@lens:~$ sudo systemctl status nagios4
● nagios4.service - nagios4
   Loaded: loaded (/usr/lib/systemd/system/nagios4.service; disabled; pre>
   Active: active (running) since Sun 2026-03-22 16:59:47 UTC; 12s ago
     Docs: man:nagios4
   Process: 15773 ExecStartPre=sh -c nagiospipe=$(sed -n "s/^command_file>
 Main PID: 15777 (nagios4)
    Tasks: 6 (limit: 2263)
   Memory: 3.4M (peak: 3.9M)
      CPU: 53ms
   CGroup: /system.slice/nagios4.service
           └─15777 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
           └─15778 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.>
           └─15779 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.>
           └─15780 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.>
           └─15781 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.>
           └─15782 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg

Mar 22 16:59:47 lens nagios4[15777]: wproc: Registry request: name=Core Wor>
Mar 22 16:59:47 lens nagios4[15777]: wproc: Registry request: name=Core Wor>
Mar 22 16:59:47 lens nagios4[15777]: wproc: Registry request: name=Core Wor>
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Mar 22 16:59:47 lens nagios4[15777]: wproc: Registry request: name=Core Wor>
Mar 22 16:59:47 lens nagios4[15777]: Successfully launched command file wor>
Mar 22 16:59:47 lens nagios4[15777]: Successfully launched command file wor>
lines 1-27/27 (END)
[6]+ Stopped                                sudo systemctl status nagios4
lens@lens:~$ |

```



Avec Zabbix

```
lens@lens:~$ sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-agent -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
zabbix-frontend-php is already the newest version (1:7.0.24-1+ubuntu24.04).
zabbix-agent is already the newest version (1:7.0.24-1+ubuntu24.04).
The following additional packages will be installed:
  fping libcares2 libevent-extra-2.1-7t64 libevent-pthreads-2.1-7t64
  libltdl7 libodbc2 libopenipmi0t64 mysql-client mysql-client-8.0
  mysql-client-core-8.0 snmpd
Suggested packages:
  odbc-postgresql tdsodbc snmptrapd virtual-mysql-server
The following NEW packages will be installed:
  fping libcares2 libevent-extra-2.1-7t64 libevent-pthreads-2.1-7t64
  libltdl7 libodbc2 libopenipmi0t64 mysql-client mysql-client-8.0
  mysql-client-core-8.0 snmpd zabbix-server-mysql
0 upgraded, 12 newly installed, 0 to remove and 1 not upgraded.
Need to get 5,487 kB of archives.
After this operation, 72.1 MB of additional disk space will be used.
Get:1 http://ht.archive.ubuntu.com/ubuntu noble-updates/main amd64 snmpd amd64 5.9.4+dfsg-1.1ubuntu3.2 [59.5 kB]
Get:2 https://repo.zabbix.com/zabbix/7.0/ubuntu noble/main amd64 zabbix-server-mysql amd64 1:7.0.24-1+ubuntu24.04 [1,777 kB]
Get:3 http://ht.archive.ubuntu.com/ubuntu noble/main amd64 libcares2 amd64 1.27.0-1.0ubuntu1 [73.7 kB]
Get:4 http://ht.archive.ubuntu.com/ubuntu noble/main amd64 libevent-extra-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [64.2 kB]
Get:5 http://ht.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7,982 B]
Get:6 http://ht.archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:7 http://ht.archive.ubuntu.com/ubuntu noble-updates/main amd64 libodbc2
```



```
no containers need to be restarted.  
User sessions running outdated binaries:  
lens @ session #3: sshd[1428]  
lens @ user manager service: systemd[1406]
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
lens@lens:~$ sudo systemctl enable zabbix-server  
Synchronizing state of zabbix-server.service with SysV service script with /  
usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-server  
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-server.se  
rvice → /usr/lib/systemd/system/zabbix-server.service.  
lens@lens:~$ sudo systemctl start zabbix-server  
lens@lens:~$ |
```

Not secure 192.168.43.79/setup.php

ZABBIX

[Welcome](#)
[Check of pre-requisites](#)
[Configure DB connection](#)
[Settings](#)
[Pre-installation summary](#)
[Install](#)

Welcome to

Zabbix 7.0

Default language English (en_US)

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Questions

1. Quelle est la différence entre logs système et logs applicatifs ?

Les logs système enregistrent les événements liés au fonctionnement de l'ordinateur et du noyau, comme les démarrages, les connexions ou les erreurs matérielles, tandis que les logs applicatifs concernent uniquement les logiciels ou services installés, par exemple Nginx ou MySQL.

2. Pourquoi centraliser les logs en entreprise ?

Centraliser les logs en entreprise permet de regrouper toutes les informations dans un point unique, ce qui facilite la surveillance, l'analyse des incidents, la sécurité et la conformité réglementaire.

3. Que se passe-t-il si les logs deviennent trop volumineux ?

Si les logs deviennent trop volumineux, ils peuvent saturer le disque dur, ralentir le système et compliquer la recherche d'informations, ce qui peut mener à des problèmes de performance ou même à une indisponibilité du serveur.

4. Quel est le rôle de logrotate ?

Le rôle de logrotate est d'automatiser la gestion des fichiers de logs en les faisant tourner régulièrement, en les compressant ou en supprimant les anciens, afin d'éviter qu'ils ne deviennent trop gros et qu'ils occupent tout l'espace disque.

5. Pourquoi surveiller la RAM et le CPU ?

Surveiller la RAM et le CPU est essentiel car une consommation excessive de mémoire ou de processeur peut provoquer des ralentissements, des plantages ou révéler une attaque (comme un déni de service), ce qui met en danger la stabilité du système.

Cocclusion.

Ce TD m'a permis de savoir comment surveiller les ressources système, analyser les logs avec journalctl, comprendre les logs applicatifs, détecter un incident et comprendre la rotation des logs, donc la tâche a été réussie.